

1.1 Use case 1

Use Case “Pay Order By Credit Card”

1. Use case code

UC233

2. Brief Description

This use case describes the process of a customer paying for an order using a credit card as an alternative to the default QR code payment method. The credit card transaction is processed through the integrated PayPal Sandbox environment via PayPal REST APIs v2. Upon successful payment, the system displays the transaction and order details , sends a confirmation email containing the invoice and transaction information , records the payment and order data , and updates the order status to pending processing.

3. Actors

3.1 Primary Actor: Customer

3.2 Secondary Actor: PayPal SandBox

4. Preconditions

- The customer has set up delivery information.
- The software displays and temporarily saves invoice information, including the list of products in the cart, quantity, product prices, total product price excluding VAT, total product price including VAT, delivery fee, and total amount to be paid.
- The software initially displays a QR code as the default payment method.
- The AIMS system is configured to connect to the PayPal Sandbox environment.

5. Basic Flow of Events

1. The customer chooses to switch from the default QR code payment method to the alternative payment method using a credit card.
2. The AIMS system initiates a payment request to the integrated PayPal Sandbox environment using PayPal REST APIs v2.
3. The PayPal Sandbox prompts the customer for their credit card information.
4. The customer enters their credit card details and submits the payment.
5. The PayPal Sandbox captures the payment and returns a successful transaction confirmation to the AIMS system.
6. After a successful payment, the software displays general information of the order (customer name, phone number, shipping address, province, total amount) and transaction information (transaction ID, transaction content and transaction datetime).
7. The software sets the order to a pending processing state.
8. The software sends invoice and payment transaction information to the customer's email.
9. The software records the payment transaction information and the successfully paid order.

6. Alternative flows

Table 1 - Alternative flows of events for UC “Pay Order By Credit Card”

No	Location	Condition	Action	Resume location
1.	At Step 1	If the customer decides to adjust the delivery method or the items they wish to purchase.	The software recalculates the delivery fees and updates the corresponding invoice.	Resumes at Step 1
2.	At any step during the payment process	If the customer chooses to go back to any previous step or exit the software.	The software halts the payment process and navigates the user back to the previous screen or exits.	Use case ends
3.	At Step 5	If the PayPal Sandbox fails to capture the payment (e.g., declined card or invalid API credentials).	The software displays an error message and does not proceed with the successful post-payment actions.	Resumes at Step 3 or Use case ends

7. Input data

Table 2 - Input data of UC “Pay order by credit card”

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Payment Method Selection	The choice to switch from the default QR code to a credit card.	Yes	Must be selected as "Credit Card" (PayPal Sandbox).	"Credit Card" (PayPal Sandbox). Credit Card
2.	Credit Card Information	The payment credentials required by the PayPal Sandbox integration to capture the payment.	Yes	Must be valid test credentials accepted by the PayPal Sandbox environment.	Sandbox Credit Card Number, Expiry, CVV
3.	Payment Payload	The total amount and	Yes	Must match the total amount to be paid.	USD 12.50

		order details sent from AIMS to PayPal Sandbox API.			
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8. Output data

Table 3 - Output data of UC “Pay order by credit card”

No	Data fields	Description	Display format	Example
1.	General Order Information	Displays the customer name, phone number, shipping address, province, and total amount.	Text	Nguyen Van A, 0987654321, 1 Dai Co Viet, Hanoi, 125,000 VND
2.	Transaction Information	Displays the transaction ID, transaction content, and transaction datetime.	Text	TXN-98765, Payment for Order #123, 2023-10-27 14:30:00
3.	Email Notification	Invoice and payment transaction information sent to the customer's email.	Email format	(Email containing the invoice and transaction details)

9. Postconditions

- The order status is updated to a pending processing state.
- The software has sent the invoice and payment transaction information to the customer's email.
- The software has recorded the payment transaction information and the successfully paid order in the system.